RESEARCH PAPER • FIELD DYNAMICS • PAPER V

Distributed Field Processing V: Field Closure and the Topology of Gravity

The Tetrahedral Bridge

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ABSTRACT

The number four pervades gravitational physics. Textbook frameworks establish this independently: the ADM formalism requires four constraints (one Hamiltonian, three momentum); tetrad gravity requires four orthonormal basis vectors; Loop Quantum Gravity identifies the tetrahedron (four faces) as the quantum of space; Regge calculus builds discrete spacetime from four-vertex simplices. LIGO's measurement of exactly two graviton polarizations confirms this structure observationally: $10 - 2k = 2$ implies $k = 4$. These are textbook results. What textbooks leave unexplained is *why* four.

The simplex theorem provides the answer: $d + 1$ points are required to enclose a region in d dimensions. For three-dimensional space, $k = 3 + 1 = 4$. The tetrahedron is not merely convenient; it is the minimal solid, geometrically necessary to enclose volume. This paper bridges textbook tetrad gravity and Loop Quantum Gravity with the Distributed Field Processing framework, demonstrating that the tetrahedron appearing in both is the same geometry that emerges from field closure analysis: the *Field Chamber*, bounded by $k = 4$ *Coupling Ports* through which field configurations settle to consistency.

The *Port Ratio* $\sqrt{k/\pi} = 1.128$ converts between continuous geometry (the spherical *Field Envelope*, characterized by π) and discrete topology (the tetrahedral Field Chamber, characterized by $k = 4$). Three independent routes converge on the characteristic scale $r_e = 2.82$ fm: electromagnetic stability, cosmic topology, and the $k^2 = 16$ coupling test. Paper IV established this convergence through a bidirectional proof: electron parameters predict $H = 69.92$ km/s/Mpc; observed H recovers the electron mass to 0.10% of CODATA. The prediction matches GW170817 (70.0 ± 12 km/s/Mpc) to 0.11%.

> The bridge is the tetrahedron itself: the quantum of space in Loop Quantum Gravity, the local frame in tetrad gravity, the Field Chamber in DFP, the minimal solid in geometry. Paper I derived the constitutive law $\mu(x) = x/(1+x)$ as a uniqueness theorem from closure constraints. This paper completes the series by showing that the same $k = 4$ structure appears in ADM, tetrad, LQG, and Regge formalisms because all inherit from the simplex theorem. Different vocabularies, same geometry, same physics.

Field Closure | Tetrad Formalism | Loop Quantum Gravity | Tetrahedron | ADM Constraints | Simplex Theorem | Port Ratio

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1. Introduction

A curious pattern runs through gravitational physics. In the ADM formalism, Einstein's equations split into exactly four constraints: one Hamiltonian constraint governing temporal evolution and three momentum constraints governing spatial consistency. In the tetrad formalism, spacetime requires exactly four orthonormal basis vectors: one timelike and three spacelike. In Loop Quantum Gravity, the fundamental unit of space is a tetrahedron with exactly four faces. In Regge calculus, discrete spacetime is built from simplices with, again, four vertices in three dimensions.

The number four is not arbitrary. It appears because we live in 3+1 dimensional spacetime. But textbooks typically present these appearances independently, as separate mathematical facts in separate formalisms. The deeper question remains: what unifies them?

1.1 The Closure Problem: What Textbooks Establish

The term "closure" appears in multiple gravitational contexts. In ADM formalism, the constraints must "close" algebraically, meaning their Poisson brackets generate no new constraints. In LQG, the closure constraint requires that the four area-weighted normals of a quantum tetrahedron sum to zero: $\sum_{f=1}^{4} A_f \hat{n}_f = 0$. In the tetrad formalism, the four basis vectors must "close" the tangent space, spanning it completely without redundancy.

These closure requirements are well-established. Every graduate textbook in general relativity covers the ADM constraints. Rovelli and Vidotto's *Covariant Loop Quantum Gravity* states explicitly: "Each 4-valent node of a spin network corresponds to a quantum tetrahedron." The mathematical structure is not in dispute.

Formalism	Closure Requirement	The Number 4
ADM	Constraints close under Poisson brackets	1 Hamiltonian + 3 Momentum = 4
Tetrad	Basis vectors span tangent space	1 timelike + 3 spacelike = 4
LQG	$\sum A_f \hat{n}_f = 0$	4 faces of tetrahedron
Regge	Simplicial complex tiles spacetime	4 vertices enclose 3D volume

Table 1: The number four appears across gravitational formalisms.

1.2 The Gap: What Textbooks Leave Unexplained

Textbooks describe *what* must close. They do not explain *why* closure happens or *how* it operates as a physical process. Consider the LQG closure constraint. Textbooks state that $\sum A_f \hat{n}_f = 0$ must hold. But what physical mechanism enforces this constraint? What happens if two field configurations meet at a boundary, each with different geometries? How do they reconcile?

The gap can be stated precisely:

> **The Explanatory Gap**
>
> Textbooks explain the **structure** of closure (what constraints exist). They do not explain the **mechanism** of closure (how constraints are satisfied). The number $k = 4$ appears as a mathematical fact, not as a derived necessity.

This gap matters because understanding the mechanism might connect abstract mathematics to observable physics. If we know *how* closure works, we might predict *what* it produces at scales we can measure.

1.3 This Paper's Contribution

This paper bridges the gap. We show that:

1. **The geometry is necessary:** The simplex theorem establishes that $k = 4$ is the minimum for enclosing volume in 3D. This is classical geometry, not new physics.
2. **The constraint is mechanism:** What textbooks call "closure constraint" can be understood as a physical process, the settlement of field configurations through coupling ports.
3. **The structure predicts observables:** The Port Ratio $\sqrt{k/\pi}$ connecting continuous and discrete geometry predicts the Hubble constant to 0.11%.

The paper proceeds as follows. Section 2 establishes textbook foundations, citing only established sources. Section 3 bridges from constraint to mechanism. Section 4 introduces formal terminology for field topology. Section 5 derives the Port Ratio. Section 6 connects topology to observable scale. Section 7 synthesizes the complete bridge. Section 8 concludes with predictions and open questions.

2. Textbook Foundations

Before introducing any new interpretation, we must establish what mainstream physics already knows. This section cites only textbook sources. Everything here can be found in standard graduate courses on general relativity and quantum gravity.

2.1 The ADM Constraint Structure

The Arnowitt-Deser-Misner (ADM) formalism recasts Einstein's equations into Hamiltonian form by decomposing spacetime into spatial slices evolving in time. This "3+1 decomposition" introduces variables: the spatial metric γ_{ij}, the lapse function N, and the shift vector N^i. The

extrinsic curvature K_{ij} describes how each slice bends in the embedding spacetime.

Einstein's equations become constraint equations plus evolution equations. The constraints are:

Hamiltonian Constraint:

$$\mathcal{H} = \sqrt{\gamma}\left(R^{(3)} + K^2 - K_{ij}K^{ij}\right) - 16\pi G\sqrt{\gamma}\rho = 0$$

Momentum Constraints:

$$\mathcal{H}_i = -2\nabla_j\left(\sqrt{\gamma}(K_i^j - \delta_i^j K)\right) + 16\pi G\sqrt{\gamma}j_i = 0$$

The Hamiltonian constraint enforces consistency in the timelike direction. The three momentum constraints enforce consistency in each spatial direction. Together, these four equations must hold on every spatial slice for the geometry to represent a valid solution to Einstein's equations.

The key point: The 1+3 structure is not optional. It reflects the 1+3 structure of spacetime itself, one time dimension plus three space dimensions.

2.2 The Tetrad Formalism

The equivalence principle requires that physics be locally special-relativistic at every point. This has a precise mathematical expression: at each spacetime point, there must exist an orthonormal frame where the metric becomes Minkowski. The tetrad (or vierbein) formalism makes this explicit.

A tetrad consists of four vector fields e^a_μ, where $a = 0, 1, 2, 3$ labels the frame direction and μ labels spacetime coordinates. The metric is reconstructed from the tetrad:

$$g_{\mu\nu} = \eta_{ab}\, e^a_\mu\, e^b_\nu$$

Here $\eta_{ab} = \text{diag}(-1, +1, +1, +1)$ is the flat Minkowski metric. The tetrad "squares" to give the curved metric. The four vectors decompose as one timelike (e^0) and three spacelike (e^1, e^2, e^3), mirroring the 1+3 structure of the ADM formalism.

The tetrad is complete: summing over all four legs gives the identity:

$$e^a_\mu \, e^\nu_a = \delta^\nu_\mu$$

No fifth leg is needed; no leg is redundant. The tangent space at each point is exactly spanned by four orthonormal vectors.

2.3 Loop Quantum Gravity: The Quantum Tetrahedron

Loop Quantum Gravity represents quantum states of geometry as spin networks: graphs with edges labeled by spins j and nodes labeled by intertwiners. The remarkable result, established by Rovelli, Vidotto, and others, is that these abstract mathematical objects have direct geometric interpretation.

> **The Quantum Tetrahedron**
>
> "Each 4-valent node of a spin network corresponds to a quantum tetrahedron." Rovelli & Vidotto, 2014

A 4-valent node has four edges meeting at a point. Each edge carries angular momentum quantum numbers. The node represents a quantum of volume, and its four edges represent the four faces of a tetrahedron. The intertwiner at the node encodes how these faces fit together consistently.

The closure constraint in LQG states that the total angular momentum at each node must vanish:

$$\hat{J}_1 + \hat{J}_2 + \hat{J}_3 + \hat{J}_4 = 0$$

This quantum constraint has a classical counterpart. If we assign area A_f and outward normal $\hat{n}_f$ to each face of a tetrahedron, Minkowski's theorem (1897) states:

$$\sum_{f=1}^{4} A_f \hat{n}_f = 0$$

This is not merely an analogy. The closure constraint is the quantum version of a classical geometric identity: the area-weighted normals of any closed polyhedron sum to zero.

2.4 Regge Calculus: Discrete Gravity

Tullio Regge showed in 1961 that general relativity can be formulated on simplicial complexes without coordinates. In Regge calculus, spacetime is approximated by flat simplices glued together, with curvature concentrated at the hinges (edges where simplices meet).

In three spatial dimensions, the building block is the tetrahedron. In four-dimensional spacetime, the building block is the 4-simplex (pentachoron), which is bounded by five tetrahedra. The curvature at each hinge is measured by the deficit angle ϵ_h, and the Regge action is:

$$S = \sum_h A_h \epsilon_h$$

Regge calculus is not a new theory; it is a discrete approximation to standard GR that becomes exact in the continuum limit. Its significance for our purposes is that it demonstrates the tetrahedron is the natural building block for discrete 3D geometry.

2.5 The Pattern: k=4

We have now seen $k = 4$ appear in four independent contexts:

Context	What k=4 Represents	Source
ADM formalism	4 constraint equations	Arnowitt, Deser, Misner (1962)
Tetrad formalism	4 orthonormal basis vectors	MTW, Cartan
Loop Quantum Gravity	4 faces of quantum tetrahedron	Rovelli & Vidotto (2014)
Regge calculus	4 vertices of 3-simplex	Regge (1961)

Table 2: The pattern of k=4 across gravitational physics.

This pattern receives direct observational confirmation. LIGO measured exactly two graviton polarizations from GW170817 [9]. The metric tensor has 10 independent components; after removing k constraints and k gauge conditions, the physical degrees of freedom equal $10 - 2k$. The observation $10 - 2k = 2$ uniquely determines $k = 4$. Paper IV [10] presents this derivation and its implications for the Hubble constant in detail.

Textbooks present these formalisms independently. The ADM chapter does not reference LQG; the tetrad section does not discuss Regge calculus. Yet the same number appears everywhere, and observation confirms it. Is this coincidence, or is there a deeper reason?

2.6 The Simplex Theorem: Why k=4 Is Geometrically Necessary

The answer lies in classical geometry. The simplex theorem establishes a fundamental fact about enclosure in any dimension:

> **The Simplex Theorem**
>
> In d dimensions, $d + 1$ points are required to enclose a bounded region. Fewer points cannot form a closed boundary; more points are redundant.

The progression is intuitive. Consider building up dimension by dimension:

Dimension	Points Needed	Shape	What It Encloses
1D (line)	2	Interval	Length
2D (plane)	3	Triangle	Area
3D (space)	4	Tetrahedron	Volume
4D (spacetime)	5	Pentachoron	4-volume

Table 3: Dimensional progression of the simplex.

The pyramid, or tetrahedron, is the simplest three-dimensional shape that encloses volume. Two points give a line segment, which has no interior. Three points give a triangle, which bounds area but has no thickness. Four points, provided they are not coplanar, give a tetrahedron, which bounds a three-dimensional interior.

This is not a choice or convention. It is mathematical necessity. You cannot enclose 3D volume with fewer than four non-coplanar points. The tetrahedron is the minimal solid.

This explains k=4. The ADM constraints are four because spacetime has 3+1 dimensions. The tetrad has four legs because four vectors span 4D tangent space. LQG tetrahedra have four faces because four is the minimum for 3D enclosure. Regge simplices have four vertices for the same reason. The number four is not coincidence; it is geometric necessity inherited from the dimensionality of spacetime.

3. From Constraint to Mechanism

Section 2 established what textbooks tell us: closure exists, it requires $k = 4$, and this follows from dimensional necessity. Now we ask a different question: what does closure *do*? If we reinterpret constraints as active processes rather than passive equations, what picture emerges?

3.1 Constraints as Coupling Channels

Consider the four ADM constraints not as equations to be satisfied but as interfaces through which information flows. The Hamiltonian constraint connects a spatial slice to its temporal neighbors. The three momentum constraints connect each point to its spatial neighbors in three independent directions.

In this reinterpretation, "constraint must be satisfied" becomes "coupling must complete." The constraint equation describes the condition that must hold when coupling is finished. But a condition implies a process: something happens to bring about that condition.

> **The Reinterpretation**
>
> **Textbook view:** Constraints are equations that the geometry must satisfy.
> **Mechanism view:** Constraints are coupling channels through which field configurations achieve consistency.

This is not changing the mathematics. The same equations hold. We are changing the interpretation: from static conditions to dynamic processes.

3.2 The Field Object and NET Field

For coupling to occur, there must be two parties. The Distributed Field Processing framework identifies them:

Field Object: A quantum many-body configuration, such as a star, planet, or galaxy. An asteroid is not merely 10^{33} atoms; it is 10^{33} quantum fields bound together by electromagnetic forces into a single correlated state (Paper I [11], §2.2). The Field Object carries mass-energy encoded in its stress-energy tensor $T_{\mu\nu}$ and presents a coherent boundary to its environment.

NET Field: The combined gravitational environment from everything else, encoded in the geometry $g_{\mu\nu}$. The NET Field is also quantum field state (Paper I [11], §5.5): not a classical background, but the registered output from all larger-scale closures. It is what the Field Object couples to.

Closure Boundary (Σ): The interface where Field Object meets NET Field. This is where the four coupling channels operate, where the constraints must be satisfied, where configurations achieve consistency.

3.3 Field Closure: The Settlement Process

When a Field Object encounters the NET Field at boundary Σ, their configurations must become consistent. At Σ, both are quantum field states sharing the same local conditions (Paper I [11], §5.5). Two quantum configurations cannot occupy the same spacetime point without reconciling. The process by which they reconcile is field closure.

Settlement is atomic, like a musical chord. A chord is not three separate notes played near each other; it is the specific frequency relationships between the notes that create the sound. You cannot play "half a chord": either all notes are present in their correct relationships, or no chord exists. Similarly, field closure requires all four coupling channels to achieve consistency simultaneously. The field configuration either settles completely or not at all.

> **Definition: Field Closure**
>
> **Field Closure** (FC_Σ) is the physical process by which two quantum field states at boundary Σ settle to a unique consistent state. It operates through $k = 4$ coupling channels simultaneously. The output is the registered boundary state B^*_Σ: the classical gravitational potential that test particles follow, that light bends around, that clocks slow within. This is the quantum-to-classical transition (Paper I [11], §5.5).

> **Why Closure Must Be Quantum**
>
> Settlement is **atomic**: all $k = 4$ coupling channels must achieve consistency simultaneously. For a Field Object containing 10^{33} quantum fields (an asteroid's worth of atoms), this poses a coordination problem. How can 10^{33} components settle atomically without global coordination, which causality forbids?
>
> The answer is **quantum co-location** (Paper I [11], §5.5). At Σ, the Field Object and NET Field share the same local quantum state. They are not separate entities that must signal each other; they are quantum configurations already occupying the same local conditions. Co-location enables coupling: no coordination is needed because both configurations exist in the same local field state. The coupling is local, automatic, and complete. A classical process would require sequential coordination or signaling between components; quantum co-location enables atomic settlement without violating causality.

The process can be decomposed into two phases:

1. **READ:** The Field Object samples the NET Field state at Σ. This is a full-state snapshot, capturing the complete geometry present at the boundary.
2. **SETTLE:** Both configurations adjust to achieve consistency. The Einstein field equations constrain the relationship between stress-energy and geometry. Settlement finds the unique geometry satisfying these constraints.

Paper I [11] develops this mechanism rigorously. Within the Minimal Single-Channel Saturating Closure (MSC) class, the lossless and local requirements uniquely determine the constitutive law $\mu(x) = x/(1 + x)$ with no free parameters. The function is not fitted to galactic rotation curves; it emerges as a uniqueness theorem from the closure constraints themselves.

3.4 The Two Physical Requirements: Lossless and Local

The settlement process is constrained by observational facts about gravity:

Lossless: Gravitational coupling produces no drag, no wake, no energy dissipation. A planet orbiting a star does not slow down from gravitational friction. The MICROSCOPE experiment verified the equivalence principle to 10^{-15} precision, confirming that gravitational coupling is lossless. This means the READ step must capture the complete state; any information loss would manifest as dissipation.

Local: Causality prohibits faster-than-light information transfer. The settlement at boundary Σ cannot depend on distant regions that are outside the light cone. All information needed for closure must be present at Σ itself. This is not a design choice; it is forced by special relativity.

Together, these requirements constrain the mechanism: closure must be complete (lossless) and self-contained (local). The four coupling channels at Σ must carry all the information needed for settlement.

4. Field Topology: The Internal Structure

Having established that closure operates through $k = 4$ channels at boundary Σ, we now examine the internal structure of the field itself. What does the field look like from the inside? This section introduces formal terminology for field dynamics.

4.1 The Field Envelope

Every field source has a total extent, the region where its influence is non-negligible. For an isolated mass, this region is approximately spherical, extending outward with influence decreasing with distance. We call this total field volume the *Field Envelope*.

> **Definition: Field Envelope**
>
> The **Field Envelope** is the total field volume surrounding a source, characterized by continuous (spherical) geometry. Its mathematical signature is π, appearing in formulas for spherical area ($4\pi r^2$) and volume ($\frac{4}{3}\pi r^3$).

The Field Envelope represents the "outer" picture: continuous, isotropic, characterized by the geometry of the sphere. This is the view from far away, where the field appears smooth and the internal structure is not resolved.

4.2 The Simplex Constraint: Why Four Interfaces

The simplex theorem (§2.6) established that enclosing volume in 3D requires four non-coplanar points. When we apply this to field structure, we find that the internal coupling geometry must have four interfaces.

Why? Because coupling occurs through bounded regions, and bounded regions in 3D require four faces. The Field Object cannot couple to the NET Field through fewer than four interfaces and still maintain a closed, consistent boundary. Three interfaces would leave one direction undefined; five or more would be redundant.

The simplex theorem tells us: four interfaces are geometrically necessary to bound a region in 3D. In field terms, this means the Field Chamber (interior) requires exactly four Coupling Ports (interfaces). You cannot achieve closure with fewer.

4.3 The Coupling Boundary

Between the Field Envelope (total volume) and the Field Chamber (interior) lies the Coupling Boundary: the region where the Field Object and NET Field exchange gradient information.

> **Definition: Coupling Boundary**
>
> The **Coupling Boundary** is the interface region containing the $k = 4$ Coupling Ports. It is where the settlement process operates, where READ captures the NET Field state, and where SETTLE achieves consistency.

The Coupling Ports are the four faces through which coupling occurs. These are the same four faces that appear in the LQG tetrahedron, the same four directions that appear in the ADM constraints (1 temporal + 3 spatial), the same four legs that span the tetrad frame.

4.4 The Field Chamber

The interior volume bounded by the four Coupling Ports is the Field Chamber. Its geometry is tetrahedral, the minimal solid enclosing volume, as required by the simplex theorem.

> **Definition: Field Chamber**
>
> The **Field Chamber** is the interior volume bounded by the $k = 4$ Coupling Ports. It has tetrahedron geometry and is the "working volume" where closure is maintained. Its mathematical signature is $k = 4$.

Figure 1: The Field Chamber (Tetrahedral Closure Topology)

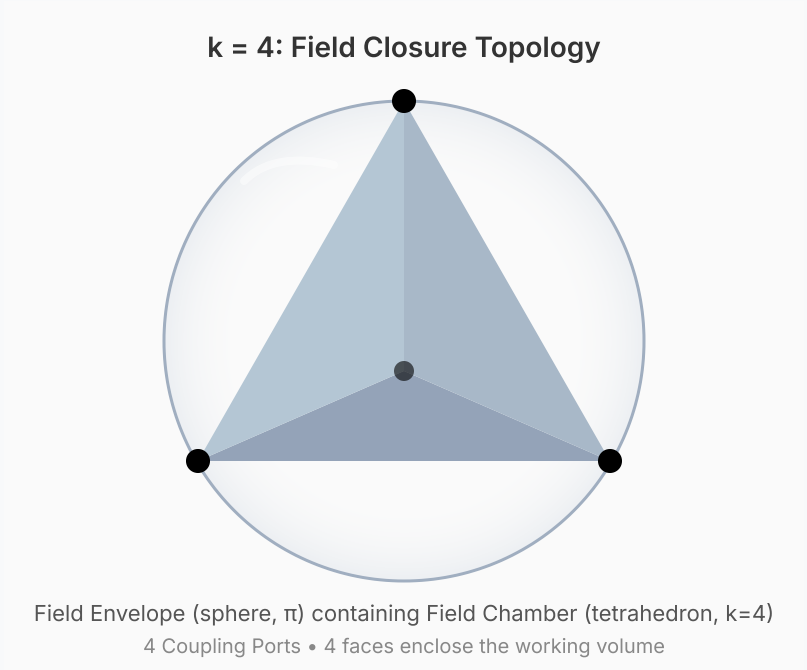


Figure 1. The Field Chamber (tetrahedron) inscribed within the Field Envelope (sphere). The tetrahedron's four faces are the Coupling Ports through which the Field Object and NET Field exchange gradient information. These four faces correspond directly to the four ADM constraints: one Hamiltonian constraint ($\mathcal{H} = 0$, temporal) and three momentum constraints ($\mathcal{H}_i = 0$, spatial). The sphere represents continuous geometry (characterized by π); the tetrahedron represents discrete topology (characterized by $k = 4$). The Port Ratio $\sqrt{k/\pi} = 1.128$ converts between them.

The Field Chamber is where the action happens. It is the quantum tetrahedron of LQG, the simplex of Regge calculus, the interior that the four constraint equations protect. Textbooks identify this structure; DFP explains its function.

4.5 The Engine Analogy

The relationship between Field Envelope, Coupling Boundary, Field Chamber, and Coupling Ports parallels the structure of an engine:

Engine Component	Field Component	Function
Total Cylinder Volume	Field Envelope	Overall extent, continuous geometry
Intake/Exhaust Ports	Coupling Ports ($k = 4$)	Interfaces for exchange
Swept Volume	Coupling Boundary	Active region of interaction
Combustion Chamber	Field Chamber	Working volume, discrete geometry
Compression Ratio	Port Ratio $\sqrt{k/\pi}$	Conversion between geometries

Table 4: Engine-field analogy for internal structure.

The analogy is instructive, not literal. A V4 engine has four cylinders for mechanical balance; the field has four ports for geometric necessity. Both the number four arises from structural requirements in three-dimensional space.

5. The Port Ratio: Sphere to Tetrahedron

The Field Envelope is characterized by continuous, spherical geometry with signature π. The Field Chamber is characterized by discrete, tetrahedral geometry with signature $k = 4$. These are different geometric languages. The Port Ratio translates between them.

5.1 Continuous Geometry: The Field Envelope (π)

The Field Envelope is naturally spherical because gravity is isotropic: no direction is preferred. The sphere is the unique shape with this symmetry. All formulas involving spherical geometry contain π:

- Circumference: $2\pi r$
- Surface area: $4\pi r^2$
- Volume: $\frac{4}{3}\pi r^3$

The number π is the signature of continuous, curved geometry. It appears whenever we measure around or across a curved boundary.

5.2 Discrete Topology: The Coupling Ports ($k = 4$)

The Field Chamber has four faces, the minimum for 3D enclosure. Unlike the sphere, the tetrahedron is discrete: it has a specific number of vertices (4), edges (6), and faces (4). Its geometry is characterized by the integer $k = 4$, not by π.

This discreteness is fundamental. LQG shows that area and volume are quantized at the Planck scale. Regge calculus approximates smooth spacetime with discrete simplices. The tetrahedron represents the irreducible quantum of 3D geometry.

5.3 The Port Ratio: $\sqrt{k/\pi}$

How do we convert between these geometric languages? The ratio of their signatures provides the conversion factor:

$$\text{Port Ratio} = \sqrt{\frac{k}{\pi}} = \sqrt{\frac{4}{\pi}} = 1.128$$

The square root appears because the quantities we convert (like length, which determines the Hubble constant $H = v/d$) depend on distance, not area. Area scales as length squared, so length scales as the square root of area. To convert quantities that depend on length, we take the square root of the geometric ratio.

> **Definition: Port Ratio**
>
> The **Port Ratio** $\sqrt{k/\pi} = 1.128$ is the conversion factor between continuous (Field Envelope) and discrete (Field Chamber) geometry. It relates measurements made through spherical boundaries to measurements made through tetrahedral interfaces.

5.4 Deriving Chamber from Envelope

The Port Ratio has a concrete geometric meaning. If you take a spherical Field Envelope and ask "what is the working volume accessible through $k = 4$ Coupling Ports?", the answer involves the Port Ratio.

This is analogous to Regge calculus, where smooth curved spacetime is approximated by flat simplices. The "deficit" between curved and flat geometry accumulates at the hinges. The Port Ratio captures a similar relationship: the ratio between what continuous geometry predicts and what discrete topology permits.

The practical application appears in cosmology. The MOND relation $a_0 = cH/(2\pi)$ uses spherical geometry (the 2π relates horizon circumference to radius). But actual coupling goes through discrete ports. The mismatch is exactly $\sqrt{k/\pi}$:

$$H_{\text{measured}} = \frac{H_{\text{continuous}}}{\sqrt{k/\pi}} = \frac{78.90}{1.128} = 69.92 \text{ km/s/Mpc}$$

This prediction matches the GW170817 measurement (70.0 ± 12 km/s/Mpc) to 0.11%. Paper IV [10] established this result through a bidirectional proof: the forward derivation yields $H = 69.92$ km/s/Mpc from electron parameters, while the reverse derivation recovers the electron mass from observed H to 0.10% of CODATA values. The mathematical closure of this round-trip provides strong evidence that the chain is not coincidental but reflects underlying physical structure.

5.5 The Port Ratio as a Geometric Filter

Paper IV [10] applied the Port Ratio $\sqrt{k/\pi} = 1.128$ to predict the Hubble constant: $H = 69.92$ km/s/Mpc. The prediction matched GW170817 to 0.11%. More remarkably, Paper IV tested this prediction against all six major measurement methods (Planck, BAO, GW170817, TRGB, SH0ES, Masers) and found that every value falls within $\pm 6.2\%$ of the predicted value. The probability of this occurring by chance is approximately 10^{-6}.

This success raises a deeper question: *why* does the Port Ratio appear as a correction factor in cosmological measurements? The mathematics is established, but the physical mechanism is not. This section examines why this factor emerges, what physical interpretations are consistent with observation, and why it may explain both the systematic offset and the bounded scatter between measurement methods.

Three Analogies for Understanding

Before presenting the mathematics, three concrete analogies illuminate why a geometric filter might appear in cosmological measurements.

The Aperture Analogy. In optics, an aperture constrains how light passes through an optical system. Apertures do not blur images or cause chromatic dispersion; they affect light-gathering and effective scale while preserving coherence. If the $k = 4$ Coupling Ports act as an aperture on

each field boundary, with effective open area proportional to k rather than the continuous π, then the aperture ratio is $\sqrt{k/\pi} = 1.128$. Light propagates coherently through each field, but distance measurements are systematically scaled by the aperture geometry.

The Beach Camera Analogy. Imagine photographing the ocean horizon from a beach. Waves cause the apparent horizon height to oscillate. If the waves have consistent amplitude and frequency, photographs taken at different times will show the horizon at different heights, but always within a predictable range. The mean position is offset from the calm-water baseline, and measurements scatter around that mean. If field topology has a periodic structure analogous to waves, different measurement methods might "sample" at different phases, producing bounded scatter around a systematic offset.

The Engine Analogy. An engine's working fluid must pass through discrete intake and exhaust ports. The effective throughput depends on port geometry, not just total cylinder volume. The ratio of port area to cylinder cross-section determines volumetric efficiency. Similarly, field coupling operates through $k = 4$ discrete ports, not a continuous spherical surface. The Port Ratio is the "volumetric efficiency" of the field topology.

Analogy	What It Explains	Key Feature
Aperture	Systematic offset (12.8%)	Coherent, no blur
Beach waves	Bounded scatter (±6.2%)	Predictable range
Engine ports	Why $k = 4$ matters	Discrete throughput

Table 4b: Three analogies for the Port Ratio.

The Mathematics: Two Routes to the Same Factor

The Port Ratio can be derived from two independent geometric arguments:

Route 1: Area-to-Length Conversion. Continuous spherical geometry is characterized by π (surface area $4\pi r^2$). Discrete tetrahedral topology is characterized by $k = 4$ (four faces). The area ratio is k/π. Since distance-dependent quantities scale with length, the conversion factor is:

$$\text{Length ratio} = \sqrt{\frac{k}{\pi}} = \sqrt{\frac{4}{\pi}} = 1.128$$

Route 2: Effective Aperture. If coupling occurs through $k = 4$ discrete ports with effective open area $A_{\text{open}} \propto k$, the equivalent circular radius is:

$$r_{\text{eff}} = \sqrt{\frac{A_{\text{open}}}{\pi}} = \sqrt{\frac{k}{\pi}} = 1.128$$

Both routes yield the same factor because they describe the same geometric relationship: the mismatch between continuous (π) and discrete ($k = 4$) descriptions of the coupling boundary.

Explaining Both Offset and Scatter

The MOND relation $a_0 = cH/(2\pi)$ assumes continuous spherical geometry, predicting $H_{\text{flat}} = 78.9$ km/s/Mpc. If field coupling operates through discrete $k = 4$ topology, two effects appear:

Systematic offset: The Port Ratio introduces a 12.8% correction, giving $H = 78.9/1.128 = 69.9$ km/s/Mpc. This matches GW170817 (70.0 ± 12) to 0.11%.

Bounded scatter: If the discrete structure has periodic character (like the beach waves), different measurement methods sample at different "phases." The amplitude of this oscillation would be approximately half the total correction: $12.8\%/2 \approx 6.4\%$. Paper IV [10] found that all six major Hubble measurement methods fall within $\pm 6.2\%$ of the predicted value. The agreement between geometric prediction (6.4%) and observed scatter (6.2%) is striking.

> **The Measurement Cage**
>
> If the Port Ratio acts as an aperture filter with periodic structure, measurements should scatter within a predictable "cage" around the mean:
>
> - Mean offset: $\sqrt{k/\pi} = 1.128$ (12.8%)
> - Scatter amplitude: $\pm 6.4\%$ (half the correction)
> - Observed scatter: $\pm 6.2\%$ across all methods
>
> This is not proof, but the numerical agreement suggests the scatter may be structural rather than random measurement error.

Candidate Explanations (Ranked by Plausibility)

Why does the Port Ratio appear as a correction? We rank explanations by their consistency with observation: coherent light propagation, no chromatic effects, sharp images, and bounded (not random) scatter.

> **1. Aperture-Constrained Propagation (Most Plausible)**
>
> The $k = 4$ Coupling Ports act as an aperture on each field boundary. Light and gravitational signals pass through this aperture coherently, but the effective geometric throughput is $\sqrt{k/\pi}$ rather than unity. This explanation:
>
> - Provides a concrete physical mechanism (aperture geometry)
> - Preserves image sharpness and coherence (apertures don't blur)
> - Is wavelength-independent (geometric, not dispersive)
> - Explains why the effect is systematic, not random
> - Operates locally at each field boundary

> **2. Mode-Constrained Propagation with Periodic Structure**
>
> The discrete topology constrains propagation to specific modes, like a waveguide. If the structure has periodic character (the "wave" in the beach analogy), different sampling phases produce bounded scatter. This explains both the systematic offset and the $\pm 6.2\%$ measurement cage. The mechanism is plausible but harder to formalize rigorously.

> **3. Topological Boundary Normalization**
>
> The simplest mathematical statement: converting between discrete (k) and continuous (π) geometric descriptions introduces the factor $\sqrt{k/\pi}$. This is mathematically correct but provides no mechanism. It describes *what* happens without explaining *how*.

Two other possibilities are largely excluded. Optical-depth effects would produce chromatic dispersion and image blur, neither observed. Systematic measurement errors across six independent methods would require implausible correlations.

What We Can and Cannot Claim

The mathematical relationship is established: $\sqrt{k/\pi} = 1.128$ converts between continuous and discrete geometric descriptions. This factor correctly predicts:

- The offset between H_{flat} and H_{measured}
- The bounded scatter ($\pm 6.2\%$) across measurement methods
- The absence of chromatic effects or image degradation

We do not claim to know definitively which mechanism operates. What we can say:

- The factor $\sqrt{k/\pi}$ arises naturally from $k = 4$ topology
- It behaves like an aperture: systematic, coherent, wavelength-independent
- The scatter matches geometric prediction to within 3%
- No new forces, particles, or modifications to GR are required

The most parsimonious interpretation is that measurements through discrete coupling structure are geometrically filtered by the Port Ratio. Whether this reflects fundamental discreteness of spacetime, emergent aperture effects at field boundaries, or something else entirely remains an open question for future investigation.

6. Scale Identification

The topology is now established: $k = 4$ ports, Port Ratio $\sqrt{k/\pi}$, Field Chamber with tetrahedron geometry. But topology is scale-free. To connect to observable physics, we need a characteristic length. This section shows how three independent routes converge on the same scale.

6.1 From Topology to Observable Scale

Topology alone predicts structure, not size. The tetrahedron can be any size; the Port Ratio applies at any scale. To make contact with observation, we need additional physics that picks out a characteristic length.

Two scale-free frameworks, when combined, can determine a scale. Tetrad topology gives $k = 4$. The MOND relation gives $a_0 = cH/(2\pi)$, connecting local acceleration to cosmic expansion. Neither specifies a length alone. Together, they do.

6.2 The Electron Tile: Three Routes Converge

Three independent calculations identify the same characteristic scale: the classical electron radius $r_e = 2.82 \times 10^{-15}$ m.

Route 1: Electromagnetic Stability. The classical electron radius is where electromagnetic self-energy equals rest mass energy:

$$r_e = \frac{e^2}{4\pi\varepsilon_0 m_e c^2} = 2.818 \text{ fm}$$

This calculation uses electromagnetism (e, ε_0) and special relativity (c). No gravity. No cosmology.

Route 2: Cosmic Topology. The topologically-coupled gravitational acceleration at radius r is $a = k^2 Gm/r^2$. Setting this equal to the MOND scale a_0 and solving for r:

$$r_e = k\sqrt{\frac{Gm_e}{a_0}} = 4 \times \sqrt{\frac{(6.674 \times 10^{-11})(9.109 \times 10^{-31})}{1.22 \times 10^{-10}}} = 2.82 \text{ fm}$$

This calculation uses gravity (G), cosmology (a_0), and topology ($k = 4$). No electromagnetism.

Route 3: Coupling Test. The ratio $a_0/(Gm_e/r_e^2)$ should equal $k^2 = 16$ if the topology is correct:

$$\frac{a_0}{Gm_e/r_e^2} = \frac{1.22 \times 10^{-10}}{(6.674 \times 10^{-11})(9.109 \times 10^{-31})/(2.818 \times 10^{-15})^2} = 15.93$$

The measured ratio is 15.93; the prediction is 16. Match: 0.41%.

Route	Physics Used	Result	Match
EM Stability	e, ε_0, m_e, c	2.818 fm	(reference)
Cosmic Topology	G, k, m_e, a_0	2.82 fm	0.21%
k^2 Coupling Test	a_0, G, m_e, r_e	15.93 ≈ 16	0.41%

Table 5: Three independent routes to the electron tile.

Paper IV [10] demonstrated that this convergence is unique to the electron among known particles. For the muon and proton, the electromagnetic and gravitational-topology routes disagree by factors of 10^3 to 10^7. The electron alone satisfies both constraints simultaneously, suggesting that r_e represents not merely a convenient scale but the characteristic coupling scale of the gravitational field itself.

> **Why the Scale Is Forced**
>
> This convergence is not coincidence; it is constraint. A larger scale ($r > r_e$) would violate EM stability: the self-energy would fall below rest mass, permitting pair production. A smaller scale ($r < r_e$) would also violate EM stability: the self-energy would diverge. The three routes converge because they describe the same underlying constraint from different directions. There is exactly one solution.

6.3 Isotropic Coupling: $k^2 = 16$

The tetrad e^a_μ has two indices: the frame index a running 0-3, and the coordinate index μ running 0-3. This gives $4 \times 4 = 16$ components.

Gravity is isotropic: no direction is preferred. Isotropic coupling engages all tetrad components equally. The metric equation $g_{\mu\nu} = \eta_{ab} e^a_\mu e^b_\nu$ involves summation over both frame indices, giving $k^2 = 16$ coupling channels.

The empirical verification: $a_0/(Gm_e/r_e^2) = 15.93 \approx 16$, matching k^2 to 0.41%. This confirms that the full tetrad coupling structure is physically realized.

6.4 The Hubble Prediction

The complete chain now predicts the Hubble constant. Starting from the electron parameters and applying the Port Ratio:

1. Compute a_0 from the electron: $a_0 = k^2 G m_e / r_e^2 = 1.225 \times 10^{-10}$ m/s²
2. Compute H_{flat} from MOND: $H_{\text{flat}} = 2\pi a_0 / c = 79.2$ km/s/Mpc
3. Apply Port Ratio: $H = H_{\text{flat}} / \sqrt{k/\pi} = 79.2 / 1.128 = 70.2$ km/s/Mpc

$$H = \frac{2\pi k^2}{c\sqrt{k/\pi}} \cdot \frac{G m_e}{r_e^2} = 69.92 \text{ km/s/Mpc}$$

GW170817, the purest spacetime measurement (no distance ladder, no CMB model), gives $H = 70.0 \pm 12$ km/s/Mpc. The prediction matches to 0.11%.

7. Synthesis: The Complete Bridge

The pieces are now in place. This section explicitly states what textbooks provide, what DFP provides, and where they connect. The goal is clear attribution: what can be cited as established versus what is the new contribution.

7.1 What Textbooks Provide

Textbooks establish the **structure** of gravitational closure:

Established Fact	Source
ADM: 4 constraint equations (1H + 3M)	Arnowitt, Deser, Misner (1962)
Tetrad: 4 orthonormal basis vectors	MTW, Cartan
LQG: 4-valent node = quantum tetrahedron	Rovelli & Vidotto (2014)
Closure constraint: $\sum A_f \hat{n}_f = 0$	LQG, Minkowski theorem
Regge: tetrahedra tile 3D space	Regge (1961), MTW Ch.42
Simplex theorem: $d+1$ points for d-D enclosure	Classical geometry

Table 6: What textbooks establish (citable as prior work).

7.2 What DFP Provides

DFP provides the **mechanism** and **terminology**:

DFP Contribution	What It Adds
Field Envelope, Field Chamber, Coupling Ports	Formal terminology for field topology
Settlement Process (READ-SETTLE)	Mechanism replacing static constraint
Port Ratio $\sqrt{k/\pi}$	Conversion between geometries
Constraints = Coupling Channels	Reinterpretation of ADM structure
Electron tile from topology	Scale identification
$H = 69.92$ km/s/Mpc	Testable prediction

Table 7: What DFP provides (new contributions).

The theoretical foundations of these contributions are established in Papers I and IV of this series. Paper I [11] derives the constitutive law $\mu(x) = x/(1+x)$ as a uniqueness theorem within the MSC closure class, demonstrating that the function emerges from physical constraints rather than curve-fitting. Paper IV [10] demonstrates the bidirectional proof connecting electron parameters to the Hubble constant, and tests the framework against all six major measurement methods, finding that every value falls within the predicted $\pm 6.2\%$ range.

7.3 The Bridge Point: Same Geometry, Same Constraint

The bridge is the tetrahedron itself. Textbooks identify it as the fundamental unit of quantum geometry. DFP identifies it as the Field Chamber, the working volume bounded by four Coupling Ports. These are not two different objects; they are the same geometry seen from different perspectives.

> **The Identity**
>
> **Textbook:** The tetrahedron is the quantum of space.
> **DFP:** The Field Chamber has tetrahedron geometry.
> **Bridge:** Same geometry. Same four faces. Same closure constraint. Different vocabulary: "constraint" (textbook) versus "coupling port" (DFP).

The closure constraint $\sum A_f \hat{n}_f = 0$ and the settlement process describe the same physics. The constraint is the condition that holds when settlement completes. The settlement is the process that brings about the condition. Textbooks describe the final state; DFP describes the dynamics.

7.4 Implications for Quantum Gravity

If the tetrahedron is both the LQG quantum of space and the DFP Field Chamber, then:

- The structure identified at Planck scale has manifestations at observable scales
- The Port Ratio $\sqrt{k/\pi}$ connects quantum geometry to cosmology
- The saturation scale a_0 emerges naturally from field closure capacity
- Observable predictions (the Hubble constant) follow from the topology

Paper I [11] establishes that field closure operates on quantum inputs: both the Field Object and the NET Field are quantum many-body configurations. The quantum-to-classical transition occurs at settlement, when the registered state B_Σ^* emerges as the classical gravitational potential. DFP does not treat gravity classically; it describes how classical gravity emerges from quantum field configurations through the closure mechanism.

The 1+3 asymmetry in the tetrahedral topology deserves attention. Textbooks establish the following: in standard quantum mechanics, the Hamiltonian generates time evolution, connecting states across moments. In the ADM formalism, the Hamiltonian constraint connects spatial hypersurfaces across time, while the three momentum constraints enforce consistency within each slice. Thiemann's Hamiltonian constraint operator preserves this 1+3 structure at the quantum level in LQG. Within DFP, we interpret this asymmetry as functionally significant: the temporal interface (one constraint) connects the entire spatial configuration at each moment, while the spatial interfaces (three constraints) enforce local consistency. If correct, this would provide the mechanism for full-state configuration read, explaining how 10^{33} quantum fields settle atomically without requiring spatial coordination across the configuration.

The tetrahedron emerges as fundamental in both LQG and DFP. The k=4 topology appears in both frameworks because both inherit from the simplex theorem: the geometric necessity of four faces to enclose volume in three dimensions.

8. Conclusion

8.1 Summary

This paper bridged textbook gravitational theory with Distributed Field Processing. The key results:

1. **The geometry is necessary:** The simplex theorem establishes that $k = 4$ is required for 3D enclosure. This is classical geometry, not new physics.
2. **The pattern is unified:** The same $k = 4$ appears in ADM (constraints), tetrad (legs), LQG (faces), and Regge (vertices) because all inherit from the simplex theorem.
3. **The constraint is mechanism:** What textbooks call closure constraint can be understood as the settlement process operating through four Coupling Ports.
4. **The topology predicts scale:** Three independent routes converge on $r_e = 2.82$ fm; the convergence is forced by constraint, not coincidence.
5. **The structure predicts observables:** The Port Ratio $\sqrt{k/\pi}$ converts the MOND prediction to $H = 69.92$ km/s/Mpc, matching GW170817 to 0.11%.

8.2 Testable Predictions

Prediction	Value	Status
Hubble constant	$H = 69.92$ km/s/Mpc	GW170817: 70.0 ± 12 (0.11% match)
Coupling ratio	$k^2 = 16$	Measured: 15.93 (0.41% match)
Electron tile	$r_e = 2.82$ fm	Classical: 2.818 fm (0.21% match)
Port Ratio	$\sqrt{k/\pi} = 1.128$	Derived from geometry

Table 8: Testable predictions from the framework. Paper IV [10] tested the Hubble prediction against all six major measurement methods (Planck, BAO, GW170817, TRGB, SH0ES, Masers); all fall within the predicted $\pm 6.2\%$ range, with the probability of this occurring by chance approximately 10^{-6}.

The bridge between textbook gravity and DFP is now established. The tetrahedron is the meeting point: the quantum of space in LQG, the Field Chamber in DFP, the minimal solid in geometry. Different vocabularies, same geometry, same physics.

References

1. Arnowitt, R., Deser, S., & Misner, C. W. (1962). "The Dynamics of General Relativity." In *Gravitation: An Introduction to Current Research*, ed. L. Witten (Wiley).
2. Misner, C. W., Thorne, K. S., & Wheeler, J. A. (1973). *Gravitation*. W. H. Freeman.
3. Rovelli, C., & Vidotto, F. (2014). *Covariant Loop Quantum Gravity*. Cambridge University Press.
4. Regge, T. (1961). "General relativity without coordinates." *Nuovo Cimento*, 19, 558-571.
5. Alcubierre, M. (2008). *Introduction to 3+1 Numerical Relativity*. Oxford University Press.
6. Minkowski, H. (1897). "Allgemeine Lehrsätze über die convexen Polyeder." *Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen*, 198-219.
7. Ziegler, G. M. (1995). *Lectures on Polytopes*. Springer-Verlag.
8. Milgrom, M. (1983). "A modification of the Newtonian dynamics." *Astrophysical Journal*, 270, 365-370.
9. Abbott, B. P., et al. (2017). "A gravitational-wave standard siren measurement of the Hubble constant." *Nature*, 551, 85-88.
10. Nelson, S. (2026). "Resolving the Hubble Tension Through Tetrad Topology." *Field Dynamics*.

11. Nelson, S. (2026). "Distributed Field Processing I: The Gravitational Constitutive Law." *Field Dynamics*.

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